Technische Hochschule Augsburg

STATISTICS

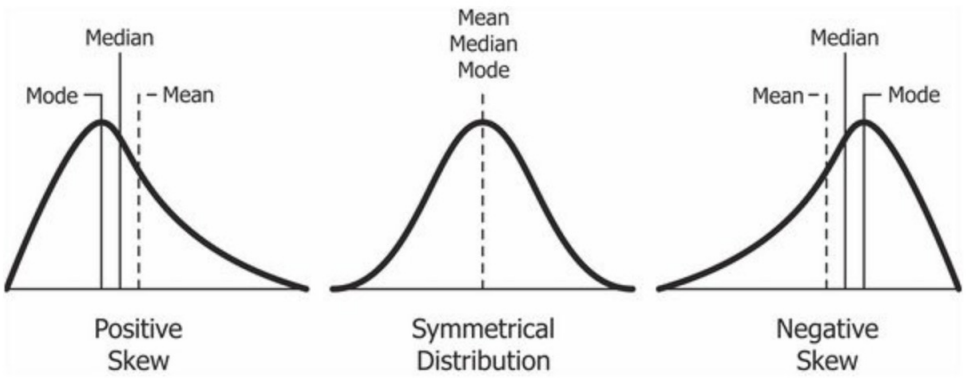
International Information Systems, 3rd semester

(4) Descriptive statistics: measures of dispersion and outliers

PROF. DR. -PHIL ALESSANDRA ZARCONE

LAST WEEK

Descriptive statistics



- The purpose of **descriptive statistics** is to summarize and visualize the data and understand patterns

- A good first step is to **plot the frequency of your data** using a **histogram**
  - Frequency distributions of data come in **different shapes** depending on their symmetry / skew, their kurtosis, the number of peaks (“modes”)

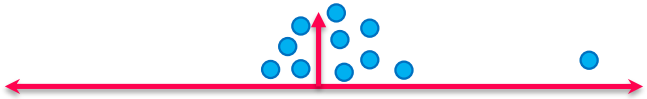
- **Mean**, **median** and **mode** are typical measures of central tendency
  - They differ with regards to the scales they can be used on, their existence in the dataset, their sensitivity to extreme values

Questions?

PRELIMINARY COURSE PLAN

- Introduction, Descriptive statistics: measures of central tendency
- **[Oct 9th-16th] Introduction to R, Descriptive statistics: measures of dispersion, distributions, visualizations**
- [Oct 23rd] Basic probability theory
- [Oct 30th] Introduction to inferential statistics: sampling, hypothesis testing
- [Nov 6th] Test Statistics (I)
- [Nov 13th] **No class *(Tag der Lehre)***
- [Nov 20th-Nov 27th] Test statistics (II-III)
- [Dec 4th] Introduction to modeling: correlation
- [Dec 11th-18th] The Generalized Linear Model
- [Jan 9th-29th] Extra session if needed, Mock Exam, QA, extra exercises

TODAY'S TOPICS AND LEARNING GOALS



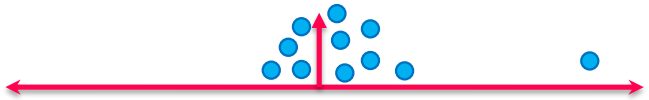
Measures of Dispersion

- compute **range**, **interquartile range**, **variance**, **standard deviation**, and understand what they tell us about the data

Outliers

- identify outliers in your data, reflect about them and decide how to handle them

MEASURES OF DISPERSION

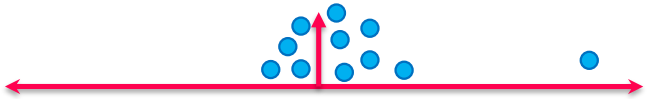


MEASURES OF DISPERSION OR SPREAD

- ***How much can we trust the values telling us about the central tendency?***
- The measures of central tendency fail to give the whole story

Claim: "this is a neighborhood with a ***high average income***"

- But mostly farmers and hourly-wage workers live in the area
- And only 3 millionaire families (weekenders)



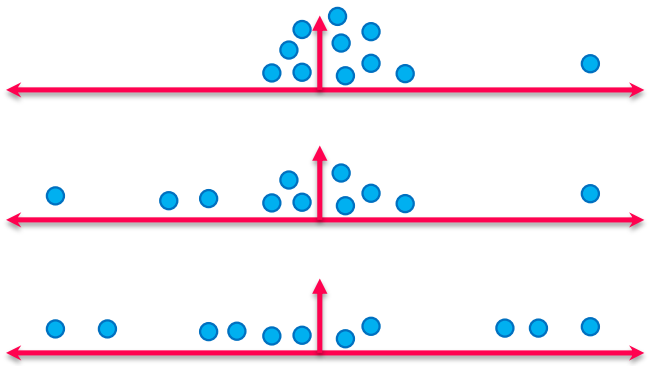
The mean income is misleading!

MEASURES OF DISPERSION OR SPREAD

- *Besides* central tendency, we need to look at the **spread** or **dispersion**, that is ***the degree to which individual data points are distributed around the mean***

Measures of dispersion:

- the ***range***
- the ***interquartile range***
- the ***variance***
- the ***standard deviation***



given the same central tendency, the dispersion of the data can vary a lot

MEASURES OF DISPERSION OR SPREAD

The Range

The easiest way to look at dispersion is the **range**

- you can obtain the **range** of the data by subtracting the smallest score from the largest score

8, 20, 29, 31, 35, 42, 55, 66, 69, 74, 195

smallest score — **largest score**

range

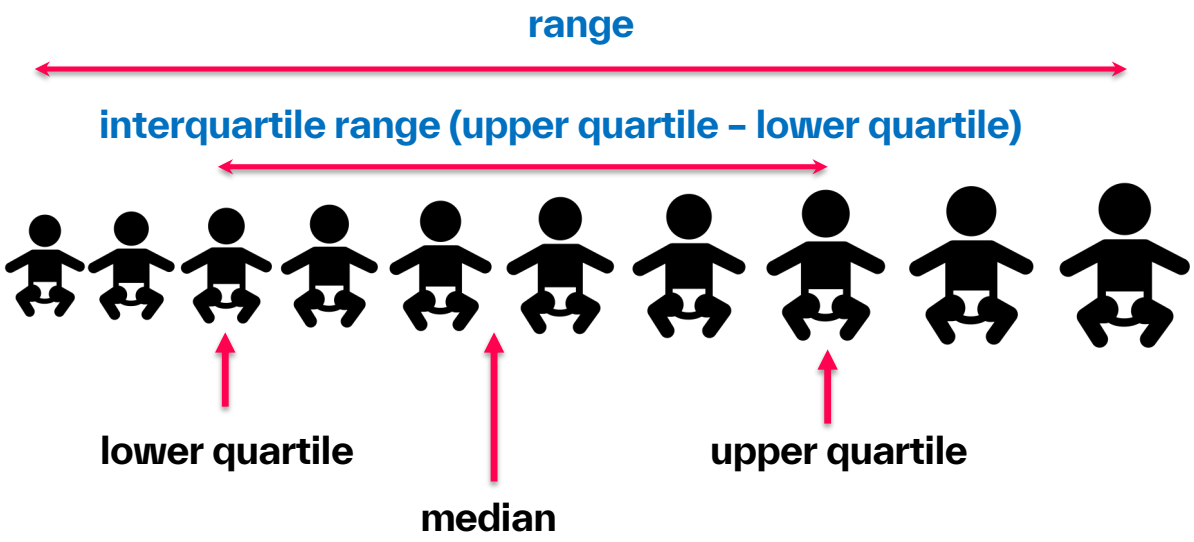
195 – 8 = 187

- a big range is probably an indication that the data is very spread out
- But: the range is affected dramatically by extreme scores
  - If we remove 195: **74 – 8 = 64**

outliers
unusually extreme values, which stand out from the rest of the distributions

MEASURES OF DISPERSION OR SPREAD

The Interquartile Range (IQR)



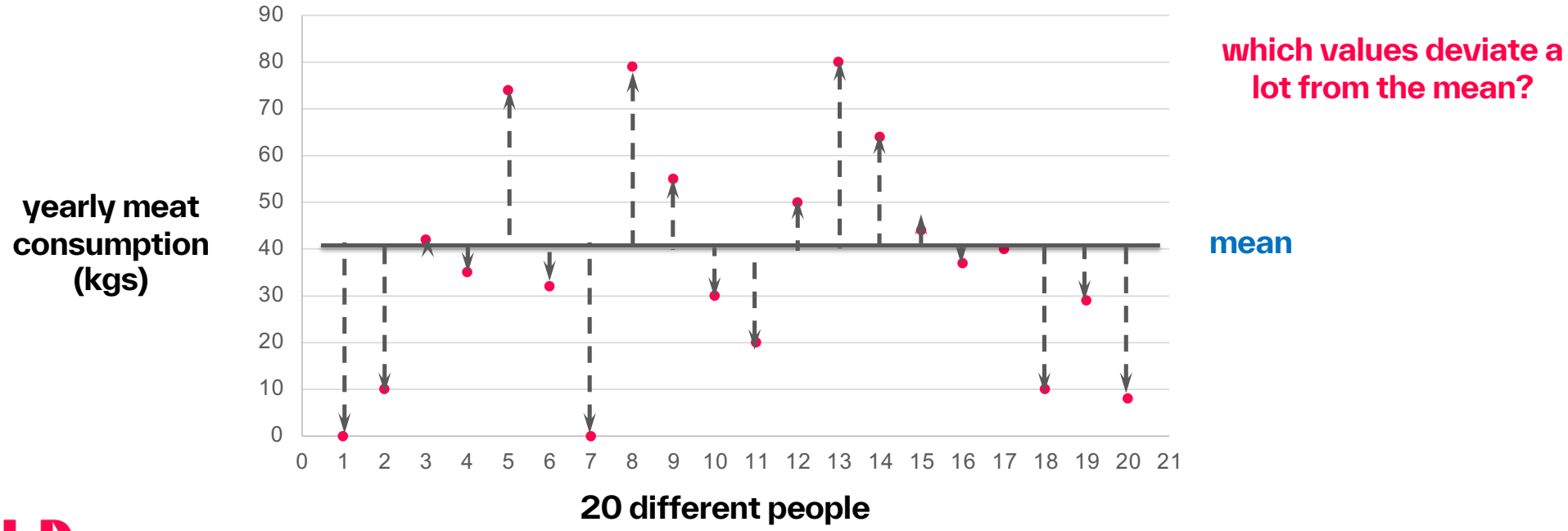
- The interquartile range ***excludes*** values at the extremes of the distribution, typically, the top 25% and bottom 25% of scores

  + it isn't affected by extreme values
  - data loss (what percentage of the scores are we looking at?)

MEASURES OF DISPERSION OR SPREAD

The Spread as Deviation from the Mean

- When looking at **spread**, we want to measure how scores are dispersed around the mean
  - That is, how much they ***deviate*** from the mean: $(X - \bar{X})$

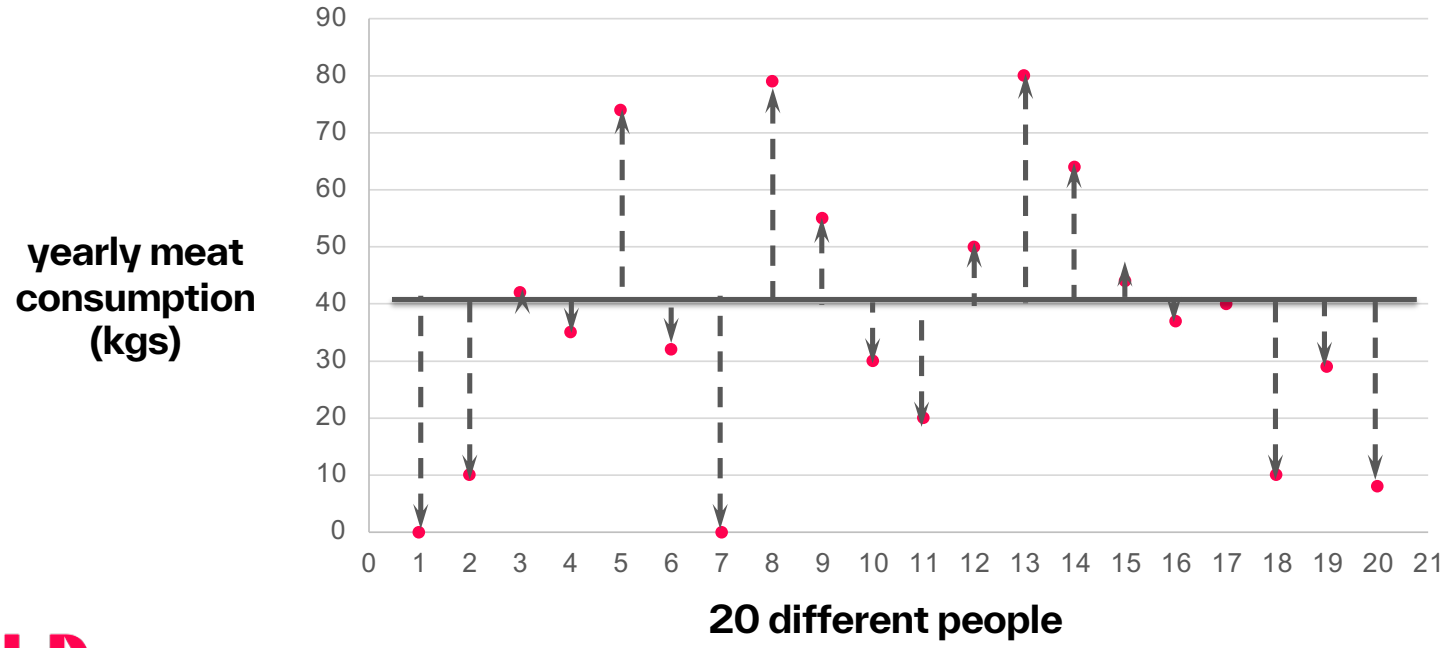


MEASURES OF DISPERSION OR SPREAD

The Average Deviation

- One way to measure how much the scores ***deviate*** from the mean is to can get all deviations from the mean and get the average over all datapoints

$$\frac{\sum(X - \bar{X})}{N}$$



But! the deviations from the mean will balance each other out!

If you don't see that, can you try using three datapoints, e.g. 1, 4 and 5?

MEASURES OF DISPERSION OR SPREAD

The Average Deviation

- An example with only 3 datapoints (1, 4, 5)

$$\bar{X} = \frac{1+4+5}{3} = 3.333$$

$$\frac{\sum(X - \bar{X})}{N} = \frac{(1 - 3.333) + (4 - 3.333) + (5 - 3.333)}{3} = 0$$

What's going on here?

- the positive and negative deviations are balancing each other out!

MEASURES OF DISPERSION OR SPREAD

The Variance

In order to make sure the deviations from the mean ***will not*** balance each other out, we can exploit the fact that a **square** of a negative number is **always positive** (very convenient!)

- We sum the **squared deviations** from the mean
- We get the average
- We use N-1 for the sample variance because it give us a better estimate of the population variance

$$s_x^2 = \frac{\sum(X - \bar{X})^2}{N - 1}$$

sample variance (computed)

$$\sigma_x^2 = \frac{\sum(X - \bar{X})^2}{N}$$

population variance (estimated)

Convention:

- Latin letters for sample statistics (measured)
- Greek letters for population parameters (estimated)

MEASURES OF DISPERSION OR SPREAD

The Variance

- An example with only 3 datapoints (1, 4, 5)

$$\bar{X} = \frac{1+4+5}{3} = 3.333$$

$$\frac{\sum(X - \bar{X})^2}{N-1} = \frac{(1 - 3.333)^2 + (4 - 3.333)^2 + (5 - 3.333)^2}{3-1} = 4.333$$

MEASURES OF DISPERSION OR SPREAD

The Standard Deviation

- The variance is based on squared deviation and thus it is not intuitively interpretable (the result is in terms of squared units)

- There's a simple solution for this: the **standard deviation (SD)** that is the positive square root of the variance
  - We can ***interpret it*** a measure of the average of the ***deviations*** of each score from the mean

We can consider values with are very distant from the mean (in terms of SD) as *outliers*

$$s_x = \sqrt{\frac{\sum(X - \bar{X})^2}{N - 1}}$$

sample SD (computed)

$$\sigma_x = \sqrt{\frac{\sum(X - \bar{X})^2}{N}}$$

population SD (estimated)

Convention:

- Latin letters for sample statistics (measured)
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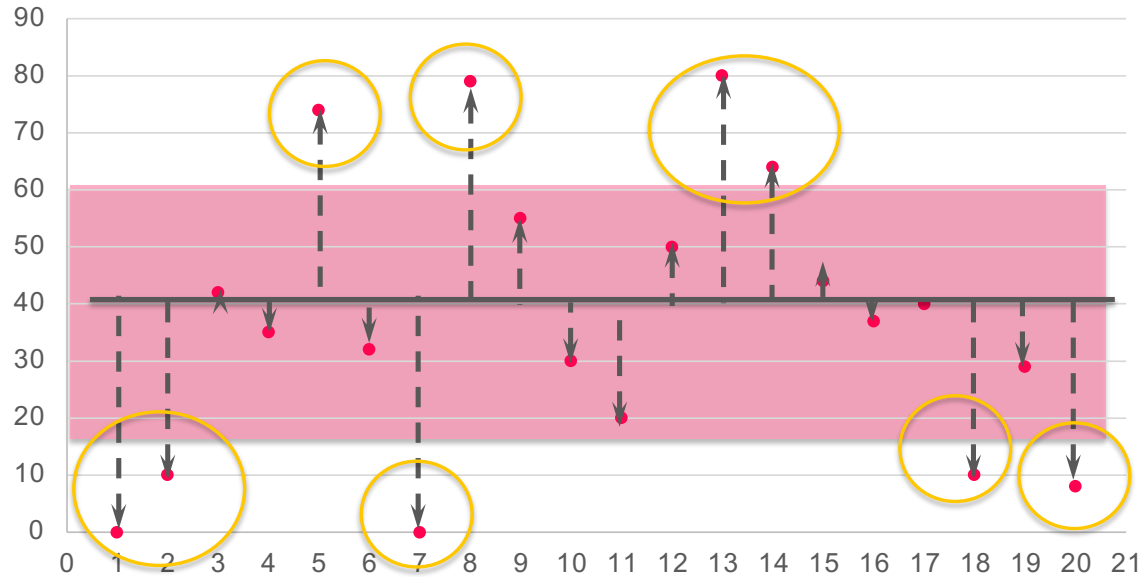
Z-SCORES

Finding datapoints more than 1 SD from the mean

↗ To understand the spread of a distribution
we can look at how many datapoints fall no more than 1 SD above or below the mean

yearly meat consumption (kgs)



20 different people

Given a datapoint, how far from the mean is it (how many SD away from the mean)?

$$\frac{\text{Score} - \text{Mean}}{\text{SD}}$$

$$\frac{X - \bar{X}}{\text{SD}}$$

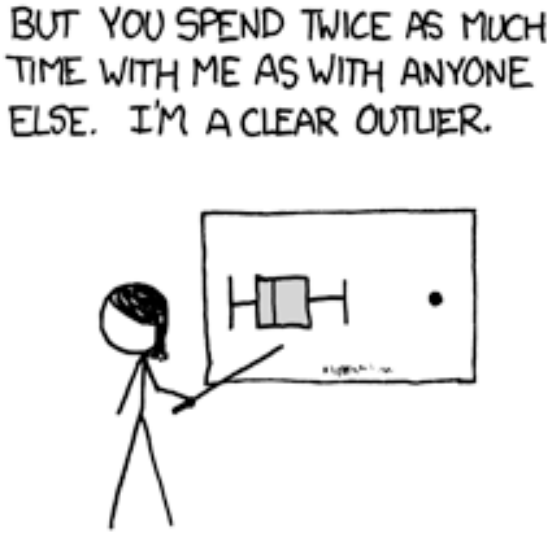
Where do I find the data falling within 1 SD from the mean?

$$\bar{X} \pm \text{SD}$$

Z-scores
indicate the deviation from the mean of a datapoint in terms of SD (how many SDs?)

Questions?

OUTLIERS


BUT YOU SPEND TWICE AS MUCH TIME WITH ME AS WITH ANYONE ELSE. I'M A CLEAR OUTLIER.

OUTLIERS

Affecting the range

The **range** of the data is affected by **outliers** (unusually extreme values)

- outliers may give the impression that the data is more spread out than it really is

8, 20, 29, 31, 35, 42, 55, 66, 69, 74, 195

smallest score — largest score

range: 195 – 8 = 187
without the extreme value: 74 – 8 = 66

The **interquartile range** of the data is less affected by outliers

- it shows not where all data can be found, but where **50% of the data** can be found

8, 20, 29, 31, 35, 42, 55, 66, 69, 74, 195

Q1 (30) — Q3 (67,5)

interquartile range: 67,5 – 30 = 37,5
without the extreme value: 63,25 – 42 = 33,75

OUTLIERS

Outlier removal: trimmed samples and trimmed statistics

- We usually want to eliminate outliers
  - if we believe they are likely to be accidents or errors
  - But we want to do so without eliminating the variability that we seek to study!

- **How can we do this?**
  - **Trimmed samples:** samples that have had a certain percentage (e.g., 10%) of the values in each tail removed
  - **Trimmed statistics:** statistics calculated on such samples

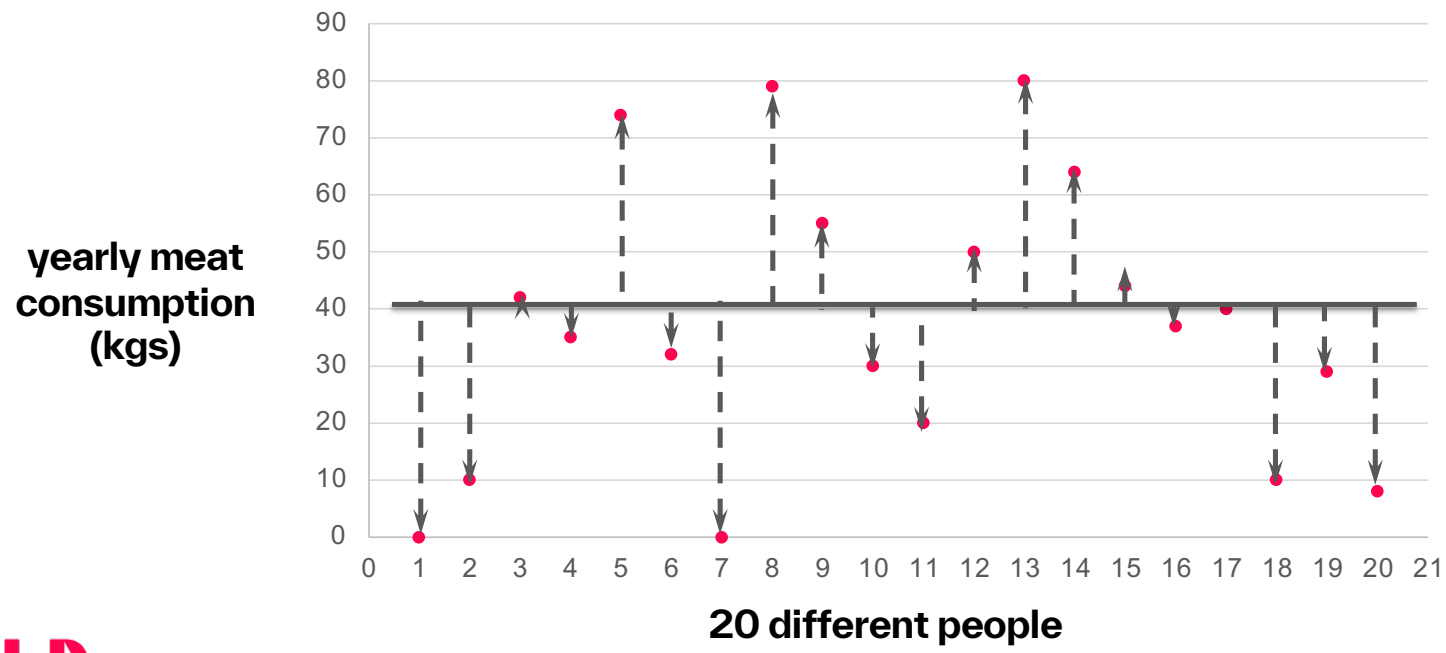
8, 20, 21, 25, 29, 31, 34, 35, 39, 42, 49, 52, 55, 59, 61, 66, 69, 71, 74, 195

20, 21, 25, 29, 31, 34, 35, 39, 42, 49, 52, 55, 59, 61, 66, 69, 71, 74

Z-SCORES

Finding datapoints more than 1 SD from the mean

- A **z-score** indicates the number of standard deviations that a datapoint is above (positive z-score) or below the mean (negative z-score)

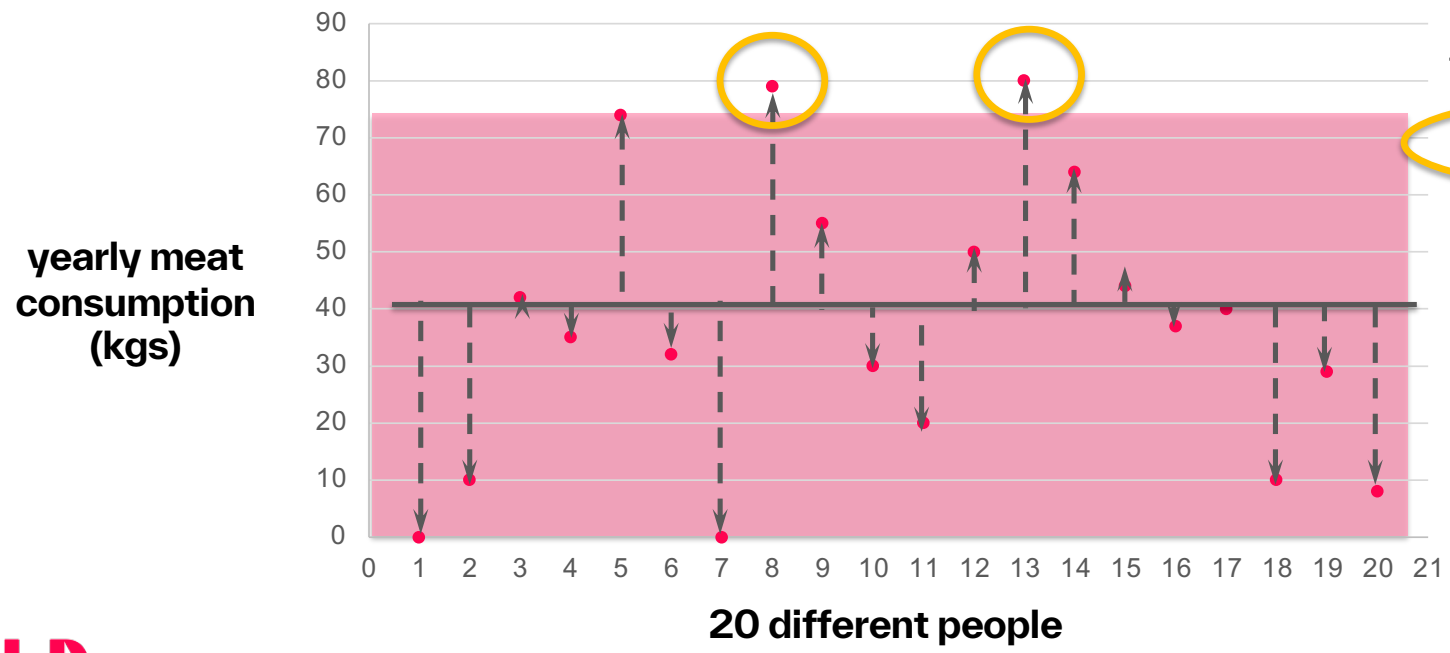


Score n	Score	Z-score
1	0	-1.48
2	10	-1.08
3	42	0.20
4	35	-0.07
5	74	1.48
6	32	-0.20
7	0	-1.48
8	79	1.69
...	...	...

Z-SCORES

Removing outliers with the z-scores

- A **z-score** indicates the number of standard deviations that a datapoint is above (positive z-score) or below the mean (negative z-score)



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...	...	...

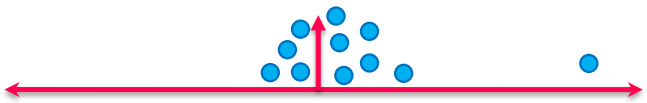
MEASURES OF CENTRAL TENDENCY WITH R

R_exercise_3.R

- In the exam, you will have to compute some measures of central tendency as well as knowing their definition / the procedure to compute them
- It's ok to use R to get **the final result**,
  but you also need to be able to provide either **the definition or the mathematical formula** if I ask you to

SUMMING UP

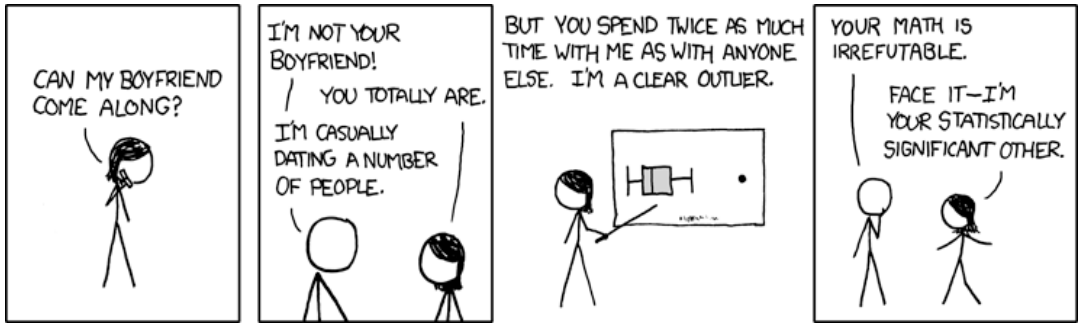
- The purpose of **descriptive statistics** is to **summarize and visualize** the data and understand patterns
- *Besides* central tendency, we need to look at the **spread** or **dispersion**, that is the degree to which individual data points ***are distributed around the mean***
  - The **range**, the **interquartile range**, the **variance**, the **standard deviation** are typical measures of dispersion
  - Looking at how much scores or datapoints ***deviate*** from the mean provides us with useful information to learn about the data spread and identify **outliers**



Questions?

HOMEWORK

Source: xkcd



- Chapter 5, sections 5.1 to 5.6 and 5.8 to 5.9 of
  Howell, "Fundamental Statistics for the Behavioral Sciences" (available on Moodle)

- Watch this videos:
  - "Measures of Spread" (Crash Course Statistics)
    https://www.youtube.com/watch?v=R4yfNi_8Kqw

- Exercises in slide 28-29

QUESTIONS AND EXERCISES

Try to do it in R, creating **R_exercise_4.R**

- There are 20 students in a dorm, and this is the number of beers they drink each weekend:

0	3	3	0	6	3	0	0	8	2
2	4	8	0	2	8	7	5	4	8

  - What's the mode? What's the median? What's the mean?

- This is the amount of meat the 20 students eat in a year (in Kgs).

0	10	42	35	74	32	0	79	55	30
20	50	80	64	44	37	40	10	29	8

  - What's the mode? What's the median? What's the mean?

QUESTIONS AND EXERCISES

Try to do it in R, continuing with **R_exercise_4.R**

- Twenty-one heavy smokers were put on a treadmill at the fastest setting. The time in seconds was measured until they fell off from exhaustion:

  18, 16, 18, 24, 23, 22, 22, 23, 26, 29, 32, 34, 34, 36, 36, 43, 42, 49, 46, 46, 57

  - What's the mode? What's the median? What's the mean?
  - What can you say about the distribution of this data?
  - What's the mode? What's the median? What's the mean?
  - Compute the mode, median, mean, upper and lower quartiles, range, interquartile range, sample variance and standard deviation
  - How many datapoints fall more than 1 SD above or below the mean? How many fall more than 1.5 SD above or below the mean?
  - What datapoints could we consider to be outliers?